

Data on the Line

Turning Demand Insights into Balanced Production · Case study · KoçDigital

1 · Meet Nimbus Dynamics

Nimbus Dynamics makes **EC fan modules** for data centre cooling systems. An EC (electronically commutated) fan is an energy-efficient fan whose on-board drive electronics can vary its speed continuously; inside there's a rotor, a stator, a shaft running in bearings, sensors and a drive board. These fans are fitted in the precision air-conditioning units of data halls and in the dry coolers that reject the servers' heat to the outside air. Nimbus Dynamics' customers are cooling-unit manufacturers and data centre builders.

Investment in AI is bringing new data halls online, and every new hall needs heat-rejection capacity, which means more fans.

Code	Name	For	Note
ND100	Standard	Precision air-conditioning units in conventional data halls	Best seller
ND200	Plus	High-density data halls	Higher airflow
ND300	Pro	Dry coolers for AI data halls	Smart fan that streams speed, temperature and vibration data; gaining share every year; its electrical and test work takes longer

How they're made. All three products come off the same assembly line. Building a fan module involves 17 tasks: preparing the housing, fitting the rotor and stator, wiring, mounting the drive board and the impeller, loading the firmware, safety and balancing tests, packing and so on. The tasks fall into four process groups: mechanical, electrical, software/test and packaging. Today they're spread over 7 stations. A unit moves on to the next station once the work at the current one is done, and the line runs two shifts a day.

Four key ideas, using Nimbus Dynamics as the example

- **Station:** a stop on the line where one or more tasks are done.
- **Takt time:** how often a unit has to come off the line to keep up with demand. It's the available production time divided by the number of units to make.
- **Line balancing:** assigning tasks to stations so that the order in which they must be done is respected and no station's load exceeds the takt time. A line can only run as fast as its busiest (slowest) station.
- **Product mix:** each product's share of what is made in a month. Because Pro takes longer to assemble, the load on the stations changes when the mix changes.

2 · What we need from you

Today is **31 December 2025**. Nimbus Dynamics' question: **can the current line meet demand for January–March 2026, and if not, what should we do?** You'll answer it in four steps:

- 1 **Demand forecast:** forecast January–March demand for the three products and show why your model can be trusted. (*due at minute 45*)
- 2 **Production plan and takt:** work out how many units have to be made each month, and the takt time.
- 3 **The line:** can the current line cope? If not, reassign the tasks to stations.
- 4 **URGENT order:** an urgent order will come in at minute 95. What do you recommend?

You'll finish with a **three-slide** recommendation, plus a cover, for Nimbus Dynamics' management. Everything you need is in the messages below.

3 · The email thread

From: Ece Demir, Project Lead · KoçDigital
To: KoçDigital project team (Nimbus Dynamics)
Date: 31 December 2025
Subject: Nimbus Dynamics · Q1 capacity review

Hi team,

Last day of the year, and we're wrapping up the Nimbus Dynamics Q1 review. We'll forecast as of today, 31 December 2025, using only what we know today.

Below is what the client sent us: Kerem's sales data and Q1 plan, and Selin's note on what's needed from the line. I've also added the WhatsApp messages Hasan, the line supervisor, sent me. They'll help you most on the line side. Read everything carefully; most of the rules for this job are in these messages.

I need your forecasts by minute 45, and they're final once they're in. Deniz from Production Planning at Nimbus will send the consensus demand plan then.

Thanks,

Ece

Forwarded message

From: Kerem Aksoy, Sales and Demand Planning · Nimbus Dynamics
Date: 30 December 2025
Subject: Sales data and Q1 plan

Hi Ece,

As discussed, here's the data: monthly sales for 2021–2025, with list prices and the average prices we actually invoiced, plus the Q1 plan. I've added the number of working days in each month too; it should come in handy.

The January Pro promotion has been approved. The March promotion for Standard is still awaiting approval; I've put it in the plan for now, and if it goes ahead, March will be strong. One more thing: Pro keeps gaining share every year, and I'd like to see that in the forecast.

Best,

Kerem

Forwarded message

From: Selin Arslan, Operations Director · Nimbus Dynamics
Date: 17 December 2025
Subject: Re: Line review

Hi Ece,

A quick note on what we need from the line. It will run with one layout from January to March; we won't change it mid-quarter.

Every additional station needs an operator on both shifts, so keep the number of stations as low as you can. If there's more than one layout with that many stations, give me the best-balanced one. After all, a line only runs as fast as its slowest station.

And then there's space: the floor has room for nine stations at most. A tenth would mean knocking down a wall.

Selin

Hasan's WhatsApp messages · 22–23 December 2025 *(translated from Turkish by Ece)*

Hasan (Line Supervisor): Hi Ece, work keeps piling up at station 5. Especially when a few Pros come through back to back. Fine on average, they say, but not at the station.

Hasan: Put it this way: when the same product comes through back to back, the operator can absorb a station time up to 20% over takt using the buffer; anything more piles up.

Hasan: And tell them not to mess with the sequence. No shaft alignment before the rotor is in, no packing before the balancing test.

Hasan: Just so they know: every task is done at the same station whatever product comes through, and we can't split a task across two stations. There's one operator per station, who does the tasks at their station in an order that respects the sequence rule. So two tasks that follow each other can share a station, as long as the later task doesn't end up at an earlier station.

Hasan: The quality audit has given us two new rules, starting in January. No station may do more than two kinds of work, and thermal compound mustn't be applied at the bench where the power terminals are wired. Station 5 breaks both right now.

Hasan: Last thing: the predictive maintenance system you set up issued an alert for the station 5 motor. It's being replaced in February, so the line will be down for two working days.

4 · Ground rules

This section is for reference: the shared definitions and expectations everyone needs so that results are calculated and handed in the same way. Which rules apply to the line is for you to work out from the messages. All data are fictional.

Timeline and deliverables

Minute	What happens
0–10	Read this document
10–45	Demand forecast
45	TeamXX_forecast.csv, downloaded by the notebook, must be uploaded to the “Minute-45 forecast” form (forms.gle/8aogMg4kcKzk1ipbA); this is the forecast that gets scored. Then a password announced in the room unlocks Deniz's consensus plan
45–57	Production plan and takt
57–90	Line balancing; meanwhile the jury asks each team a few questions
90–95	Finish the line design and the line slide
95	A second password unlocks the URGENT order
95–115	URGENT order and final touches to the slides
115	Three files must be uploaded to the “Final submission” form (forms.gle/zkj9uEyNLHyWTqFs9): TeamXX_submission.zip (containing forecast.csv, answers.csv, assignment_main.csv and, if needed, assignment_urgent.csv), your deck (TeamXX_slides.pptx, a cover and three slides) and your notebook (TeamXX_notebook.ipynb). Nothing is accepted after this point
115–120	Short closing: actual demand and the automatic points. The full closing deck and the final ranking are shared the same day

The session lasts 120 minutes in all and hand-in closes at minute 115, so don't leave the deck until the end. From Part 2 of the notebook (production plan) onwards, one person should fill in the slides alongside the analysis: the forecast and plan slide by minute 57, the line slide by minute 95, leaving the URGENT slide and final touches for minutes 95–115. Don't leave the uploads to the last minute: upload as soon as you're ready and upload again if you need to.

Type your team number at the top of the notebook (as in TEAM = "Team03"); your file names start with it. There are two forms: *Data on the Line – Minute-45 forecast* (forms.gle/8aogMg4kcKzk1ipbA) and *Data on the Line – Final submission* (forms.gle/zkj9uEyNLHyWTqFs9); your instructor also shows the links on screen. Sign in to the forms with the Google account you use for Colab. If you upload a file again, your latest upload within the time limit counts.

You'll work in Case_Notebook.ipynb; Colab is the easiest place to run it. The notebook loads the data, unlocks the protected files, checks your work and writes the submission files. The modelling is yours.

Definitions

- **Backtest:** our team's standard practice is to use eight forecast origins (an origin is the last month with known data), one every three months from December 2023 to September 2025, each forecasting the following three months; from the December 2023 origin, for example, you forecast January–March 2024. Benchmark the 72 forecasts (8 origins × 3 months × 3 products) against the seasonal naive method (the same month last year).
- **Error measures:** $WAPE = \sum |Actual - Forecast| / \sum Actual$; $Bias = \sum (Forecast - Actual) / \sum Actual$.
- **History:** in 2021–2025, promotions, prices and the calendar were fixed at least three months before the month they applied to, and the average invoiced price follows that price list. You can therefore use those values in the backtest.

- **Available time (s)** = [working days × shifts × (shift length – break) – working days × daily maintenance – additional stops] × 3,600. All time inputs are in hours.
- **Product mix:** each product's share of planned production in the month you design the line for. The notebook also tests your line in every month with that month's own mix.
- **Weighted task time:** a task's average time across the product mix. Multiply each product's time for the task by that product's share of the mix, and add the results.

Calculation notes

- Don't round the stock targets along the way. Round each product's monthly production up to a whole unit only at the last step.
- Units that fail the test and are scrapped still go through every task. Account for the yield loss in the number of units; don't apply it to the task times as well.
- Work out any limit you derive from the takt time from the unrounded takt. Keep full precision in capacity calculations too; you can round the numbers in your tables for readability.
- Every open station must have at least one task. You don't have to keep the current layout, and moving a task to another station costs nothing.
- Transport times, the order in which products enter the line, operator skills and new equipment are out of scope.

What we expect from your model and deliverables

- Design the line with a **mixed-integer linear programming (MILP)** model: the objective and the constraints must be linear in the decision variables. You can use binary (0–1) and integer variables.
- Use a **single objective function**.
- Check the layout you find in each of the three months, with that month's own product mix and available time; the notebook does this for you. If the layout fails in any month, rethink your design, and say so clearly in your slides.
- If you can't get your model to work, the notebook has a ready-made heuristic. It only looks at the sequence and the takt and knows nothing about the rules in the messages, and there's no guarantee it finds the best line. Automatic scoring looks at the line you hand in (whether it is feasible, how many stations it has and how heavily loaded the busiest station is), not at how you found it; the jury, however, may ask you to explain your constraints and your method.
- Enter your numerical answers in the answers dictionary in the notebook's last cell, which saves them to `answers.csv`: the takt time for each of the three months, your design month, its planned production, product mix and lower bound on the number of stations, the number of stations you propose, the load of the busiest station on the current line, the takt time and the lower bound on the number of stations after the URGENT order, and your decision on the URGENT order. If you decide to prebuild, also give the number of units to build early in January.
- The forecast that gets scored is the forecast file you upload to the form by minute 45. Changing your forecast after the consensus plan comes out won't change your score.
- In your slides, briefly explain your assumptions and modelling choices alongside your recommendation.

Data files (data/)

File	What it covers	Key columns
demand_history.csv	Monthly sales 2021–2025 (60 months × 3 products)	demand, promotion flag, average invoiced price, list price, working days, calendar note

File	What it covers	Key columns
promo_price_plan.csv	The January–March 2026 plan Kerem sent	promotion, approval status (Promo_Status), planned price, working days
production_calendar.csv	The January–March working calendar	working days, shifts, breaks, daily maintenance, additional stop hours
products.csv	The three products	stock on 31 December, first-pass yield (share of units that pass the test first time)
tasks.csv	The 17 assembly tasks	task name (Task_Name), process group (Process_Group), time per product (s)
precedence.csv	Which task has to come before which	earlier task, later task
current_line.csv	Today's layout: which task is at which station	task, station
*_template.csv	Submission templates	—

Scoring (100 points)

Section	Points
Forecast: file, accuracy, method and information used	25
Production plan and takt time	20
The line: valid line, number of stations, busiest station, load of the busiest station on the current line	30
URGENT order	10
Presentation	15